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Multiphysics Design Methodology for High-Speed Synchronous Reluctance Machines

Arash Allahyari, *Graduate Student Member, IEEE,* Ilya Petrov, Juha Pyrhönen, *Senior Member, IEEE,* Juho Montonen, Lassi Aarniovuori, *Senior Member, IEEE*

***Abstract*— This study introduces a multi-physics design methodology to enhance the performance of high-speed synchronous reluctance machines (SynRMs). The primary innovation is to optimize the design of the rotor flux guides (FGs), instead of the commonly optimized flux barriers. This may sound like a minor change and just a dual model, but the difference in results is significant and allows different mechanical solutions. In case of transversally laminated rotor targeting high speeds requires thick radial and tangential ribs to support the flux guides, but the ribs diminish torque and power factor and negatively impact efficiency. Reducing the rotor mass can decrease the thickness of these ribs, improving performance.**

This research combines the optimization of rotor FGs with effective mechanical support strategies based on the machine's maximum speed, comparing two constructional methods: conventional radial and tangential ribs, and non-magnetic backplates between flux guides. Optimizing the rotor geometry with FGs provides deeper insight into the design process and ensures the rotor is electromagnetically and mechanically optimized for maximum performance. To demonstrate the effectiveness of this methodology, a 6-pole-36-slot motor with 300 kW peak power featuring 2, 3, or 4 FGs, identical stator dimensions, and a maximum speed of 12,000 r/min is designed, optimized, and compared. A prototype of a 6-pole-36-slot motor with 3 FGs manufactured with backplates is presented, along with initial measurement results.

***Index Terms*— Bridgeless rotor, ribless rotor, Backplate technology, High power traction, Optimization, Synchronous reluctance machines.**

I. INTRODUCTION

CURRENT trends in traction motors can be summarized to two categories, increasing operating speed of the traction motors to reduce material and cost of the motor and reduction or elimination of rare-earth materials (magnets) due to environmental and logistic risks. However, reaching these goals brings complications at the design level and requires proposing new technologies. Synchronous reluctance machines (SynRMs) as a magnet-free motor can be a good candidate to be investigated to fulfill the goal of fully removing rare-earth magnets from traction motors.

Currently, majority of traction motors in electric vehicles are interior permanent magnet motors (IPMs) due to their capability to deliver high performance and high efficiency with the help of strong magnets that include rare-earth material.

Reducing or eliminating these magnets leads to decrease in motor power output and efficiency. Traditionally, reducing or removing magnets would require increasing the machine’s size to maintain the same torque level. However, this torque reduction can be partially or fully compensated by increasing the motor's operating speed to keep the same power level. Issues may arise due to high speeds and high excitation frequency. Namely, rotor should be mechanically strong enough to withstand centrifugal forces, higher copper and iron losses can happen due to high excitation frequency and complicate the thermal management of the motor. Therefore, the design of high-speed motors requires considerations from electromagnetic, mechanical and thermal aspects. Mechanical and thermal constraints can influence the electromagnetic design and optimization of high-speed machines in different ways, depending on the type of machine. For example, the arrangement of flux guides’ mechanical support has a clear impact on the rotor magnetic circuit design. Consequently, it is imperative to implement an effective optimization methodology to investigate and reach a good design.

Basically, one challenge for all high-speed rotors is ensuring sufficient mechanical strength to withstand the centrifugal forces generated at high rotational speeds [1]. Furthermore, certain high-speed rotors, such as solid rotors in induction machines (IMs), can generate significant amount of losses in the rotor and thereby, thermal issues can arise in such configurations that can affect stator operating temperature as well [2,3].

Moreover, the rotor structure of high-speed PMSMs and IPMs require special designs and considerations such as using

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A. Allahyari is with the Department of Electrical Engineering, Lappeenranta-Lahti University of Technology, 53850 Lappeenranta, Finland (e-mail: arash.allahyari@lut.fi).

I. Petrov is with the Department of Electrical Engineering, Lappeenranta-Lahti University of Technology, 53850 Lappeenranta, Finland (e-mail: ilya.petrov@lut.fi).

J. Pyrhönen is with the Department of Electrical Engineering, Lappeenranta-Lahti University of Technology, 53850 Lappeenranta, Finland (e-mail: Juha.Pyrhonen@lut.fi).

L. Aarniovuori is with the Department of Electrical Engineering, Lappeenranta-Lahti University of Technology, 53850 Lappeenranta, Finland (e-mail: Lassi.Aarniovuori@lut.fi).

J. Montonen is with the Danfoss Editron, Lappeenranta, Finland (e-mail: juho.montonen@danfoss.com).

carbon fiber rotor sleeve or thick radial and tangential ribs to reduce mechanical stress to an acceptable range. Furthermore, due to the existence of magnets, utilizing a rotor cooling method to ensure safe operating temperature of the magnets is crucial [4-7].

In principle, the rotor of SynRMs is simpler than that of IPMs or induction machines and is comprised of magnetic and nonmagnetic parts. Magnetic parts are steel material, and nonmagnetic areas can be air or nonmagnetic solid material.

There are two types of laminations for rotor construction of SynRMs, axially and transversally laminated rotors. Axially laminated rotors (ALR) have axial magnetic and nonmagnetic layers bonded to each other. Transversal lamination (TL) is the conventional method utilized for most of the motor laminations such as in SynRMs where thin round steel laminations are connected together with backlack resin, interlocking or welding or just with suitable mechanical means in the axial direction. TL-rotor laminations are popular because they can be manufactured using the same tools as stator laminations.

ALRs, however, require different and maybe more complicated manufacturing methods. Mostly, rotors with two poles are prototyped. A two-pole ALR with 12 kW power and 24,000 r/min speed with rotor constructed of 47 layers of S355J0 and Inconel 718 using hot isostatic pressing or vacuum brazing is investigated in [8].

Design of SynRMs requires careful optimization of the rotor and stator structure based on performance indices such as torque, torque ripple and power factor at multiple operating points. The design of the rotor is as important as the design of the stator because shape of flux guides (flux barriers) should be adjusted carefully to ensure the highest possible saliency as well as reducing torque ripple to minimum possible value [9-11].

An analytical approach aimed at simultaneously minimizing torque ripple and maximizing average torque by separating related variables presented in [12, 13]. Results in [12] and [13] demonstrate that average torque can be optimized by selecting the appropriate insulation ratio, while torque ripple can be minimized by adjusting the rotor slot pitch along d-axis. This method simplifies the rotor design of SynRMs. However, in the proposed method effect of stator design is not considered effectively, thereby, results obtained from this approach can be affected if stator dimensions or slot pole combination is changed.

The rotor of SynRMs is typically made entirely of electric steel material. As a result, rotor cooling is often unnecessary, since the rotor of a SynRM can withstand higher temperatures than other rotor types. However, the mechanical strength of the rotor is of high significance for high-speed applications and mechanical strength consideration highly affects both electromagnetic performance and manufacturability. Mechanical stress analysis of the rotor at high speeds is crucial. Mechanical and electromagnetic design of a 6-pole, 36-slot SynRM with power of 75 kW and maximum speed of 12 000 r/min for hybrid electric vehicles is presented in [14].

In TLA rotors high speeds require thicker ribs, and this is the main reason behind performance deterioration in high-speed SynRMs which corresponds to increase of quadrature inductance due to required thicker ribs. Since the d-axis inductance value remains about the same, the increase of q-axis inductance results in lower saliency. Hence, removing radial and tangential ribs is important. For instance, ALRs do not need these ribs, hence, a higher saliency can be achieved. However, production of ALRs require special processes and production of higher than two poles is much more complicated.

A superior approach to reducing the adverse effects of radial and tangential ribs could be to have nonmagnetic ribs. In this regard, dual phase (DP) materials are investigated in [15] and [16]. These DP materials require special treatment to make ribs nonmagnetic after cutting. However, they have lower relative permeability (~1059), lower magnetic saturation point (1.5T) and higher resistivity than conventional electric laminations. These features result in lower torque and higher losses making this approach impractical. Furthermore, the yield strength of these materials is reported to be roughly 500 MPa [16]. [17] provides analysis of DP material in case of a 90 kr/min SynRM. A mechanical stress analysis was provided to figure out the thickness of the ribs.

Furthermore, utilization of adhesive resin in the rotor flux barriers to hold the rotor together is investigated in [18] for 700 W, 6000 r/min SynRM. However, the adhesive material itself increases the mass of the rotor pole and causes additional centrifugal forces. Hence, this approach applied to a small motor and is challenging in case of high-power high-speed application in range of 200 kW.

This study proposes an optimization methodology for high-speed rotor of SynRMs based on torque production capability, torque ripple and effective rotor mass. Effective rotor mass is a key optimization objective to minimize the rotor's mass per pole. This approach is highly beneficial for high-speed applications since the main bottleneck is the mechanical integrity of the rotor. Moreover, this study proposed utilizing non-magnetic backplates instead of conventional ribs to support flux guides. To verify the effectiveness of this approach a comprehensive performance comparison between rotors with ribs and backplates are provided.

Patents related to eliminating bridges for SynRMs are summarized. [19,20] from LG propose bolts in axial direction to hold the outermost FG against centrifugal forces. In [21] support plates were suggested to hold the FGs together using radial bolts from both sides of support plates. Moreover, to further strengthen the rotor an axial bar (like dog bone structure) in middle of q axis from outermost FG is suggested [21]. In another patent using rotor sleeve is suggested by Edison Innovations while having the thinnest possible bridges [22].

This paper is organized as follows, firstly, optimization methodology is explained in detail in section II. Section III provides results of optimization for the rotors with 2, 3 and 4 FGs. Section IV is dedicated to mechanical stress analysis to calculated thickness of ribs. Section V provides results of 3D finite element analysis (FEA) to calculate the performance of rotors with backplates and a comprehensive comparison with bridged rotors are presented. Section VI demonstrates the manufactured motor with no load measurements.

II. PROPOSED METHODOLOGY FOR MULTIPHYSICS BASED DESIGN OF SYNRMS

The synchronous reluctance motors have attracted attention due to their simpler structure of the rotor in comparison to interior permanent magnet motors (IPMs) which are widely used in traction motors. IPMs can reach higher peak torque and efficiency values in comparison with SynRMs. However, in IPMs in flux weakening regions some current is used to weaken the flux of the magnets to keep the motor operating at the voltage limit and this additional current usage causes additional copper and iron losses in flux weakening region. The actual amount of these additional losses is highly affected by design of the IPM, operating points and control method (MTPA, loss minimization, power factor maximization etc.). However, flux weakening in SynRMs is simply achieved by decreasing the excitation current and adjusting current vector angle to keep the motor under voltage limit. Decrease in current causes the torque and power to drop but there is no additional loss created in the flux weakening region and hence, it does not have a negative effect on the efficiency of the motor. This is also a good characteristic from the thermal point of view for the motor. Furthermore, IPMs create losses in no load operation at high speeds while SynRMs do not have that kind of no-load losses. These no-load losses can happen while vehicles are cruising on highways. Hence, in theory a well-designed SynRM can reach efficiency values as high as IPMs in some regions of efficiency map.

This research aims to provide a methodology to design highly efficient high-speed synchronous reluctance motors (SynRMs) with focus on traction applications and demonstrate the capabilities of the SynR machines at a power as high as 300 kW peak and maximum speed of 12,000 r/min and investigate backplate technology to enhance the performance of SynRMs. The specifications of the SynR machine under study are described in Table I.

The original idea of this paper is to effectively use the steel material in the rotor (optimize rotor geometry) to improve the performance of SynRMs and minimize the mass of rotor so that a given machine can operate at higher speeds without the need to strengthen the rotor mechanically. The proposed methodology as described graphically in Fig. 1 comprises several steps of electromagnetic and mechanical designs.

Mechanical analysis is performed after electromagnetic optimization is completed due to several reasons. Firstly, for a rotor without ribs electromagnetic (EM) optimization can be made reliably without mechanical analysis as the backplate thickness can be selected afterwards based on the stresses. Moreover, required high number of iterations in electromagnetic optimization, integration complexity of combining electromagnetic analysis with stress analysis and most importantly slower analysis capabilities in structural stress calculations while in Ansys Maxwell several FEA models can run simultaneously which accelerates the optimization process significantly. It should be noted that consideration of effective rotor mass in objective function ensures that mechanical considerations are indirectly included at EM optimization.

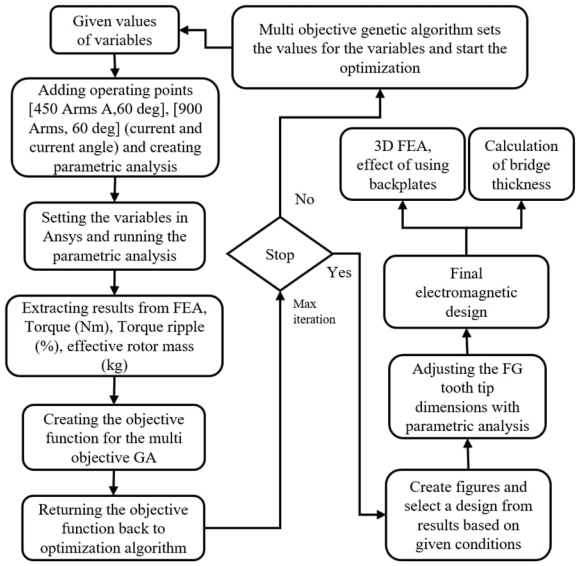


Fig.1, The proposed optimization/design methodology

TABLE I
SynR machine specifications and stator dimensions [23]

Design variables	value
No. of stator slots, Q_s	36
No. of poles, $2p$	6
No. of phases, m	3
Nominal power	200 kW
Nominal torque	550 Nm
Nominal speed	4000 r/min
Peak power	300 kW
Maximum speed	12,000 rpm
Stator outer diameter, D_{se} [mm]	375
Stator inner diameter, D_s [mm]	209.3
Stator teeth width [mm]	10
Airgap, δ [mm]	0.8
Stack length, l [mm]	210
Stator steel material	M270-35A
Rotor steel material	S355
Backplate material	Austenitic stainless steel 316L
Cooling method	Water jacket
Winding arrangement	distributed, integral-slot, 5/6 short pitch

Firstly, comprehensive optimization is performed to electromagnetically design the rotor with objectives such as average torque, torque ripple percentage and rotor effective mass. The effective mass of the rotor is created by the parts of the rotor that are not in direct connection with the rotor shaft. Moreover, the effective mass refers only to the mass of outer FGs in the rotor without considering the mass of backplates or the mass of radial and tangential ribs. Then, a design is selected from the results created. Moreover, the geometry of FG tooth tips is adjusted to further reduce the torque ripple and increase the average torque. Since FG tooth tips have two dimensions, consequently, a simple parametric analysis would be sufficient to determine the dimensions. Lastly, a mechanical analysis is performed to calculate stress and decide the thickness of ribs to hold the rotor at maximum speed of 12,000 r/min.

Furthermore, this research proposes using backplate technology to fix the FGs in the rotor construction instead of

radial and tangential ribs and investigates the effect of adopting such an approach in the performance of the machine. Ultimately, the performances of conventional transversally laminated rotor and the backplate rotor are compared.

III. OPTIMIZATION OF ROTOR WITH 2, 3 AND 4 FLUX GUIDES

To investigate and demonstrate the capabilities of SynRMs thoroughly, three different rotors have been introduced and optimized. The stator construction and operating points (current and current vector angle) have, however, remained the same. The rotors are designed with 2 FGs, 3 FGs and 4 FGs. Specifications of the machine and stator dimensions are summarized in Table I.

Optimization is implemented considering continuous and peak current values 450 Arms and 900 Arms (6 A/mm^2 and 12 A/mm^2 current densities) with the same current vector angle of 60 degrees as illustrated in Fig. 1. Under load, the SynRMs studied here in practice produce the highest torque with 60 degrees current vector angle. The models do not have any tangential or radial ribs. For the final design, based on results of mechanical stress analysis, the thickness of radial and tangential ribs will be calculated. Moreover, using 3D FEA the effect of using non-magnetic backplates to hold the rotor FGs together will be investigated in section V.

The optimization variables for a given flux guide (FG) and FG tooth tip are shown in Fig. 2 and defined in Table II. There are five variables for each FG. Consequently, there are 10 variables for the rotor with 2 FGs, 15 variables for 3 FGs and 20 variables for 4 FGs.

Results of optimization using multi objective genetic algorithm are presented in Fig. 3 for three different rotors.

The objective function includes:

- Maximizing the sum of torque at the two operating points
- Minimizing the sum of torque ripple (peak to peak divided by average torque) at two operating points
- Minimizing the effective mass of the rotor

Fig. 3 shows the results of optimization for the three rotors. Three figures are presented for optimization of each rotor namely, average torque versus effective rotor mass, average torque ripple versus effective torque and 3D map of torque vs. torque ripple vs. effective rotor mass.

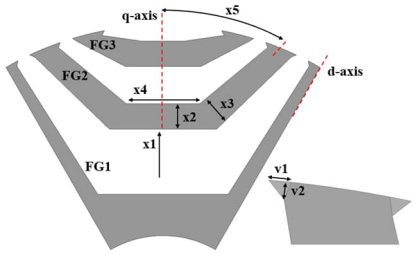


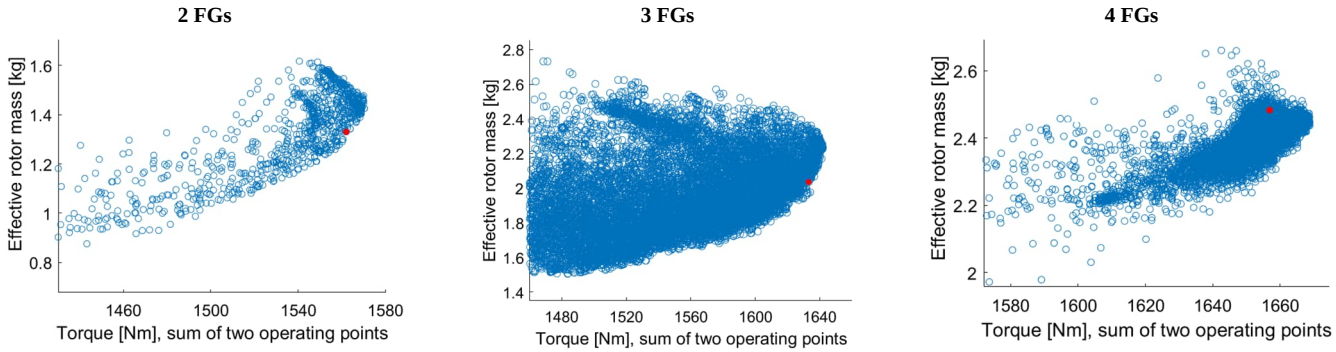
Fig. 2, Variables for a given flux guide and flux guide tooth tip, ideal case rotor (w/o ribs, w/o backplates)

TABLE II, definition of variables for each flux guide (FG) based on Fig. 2

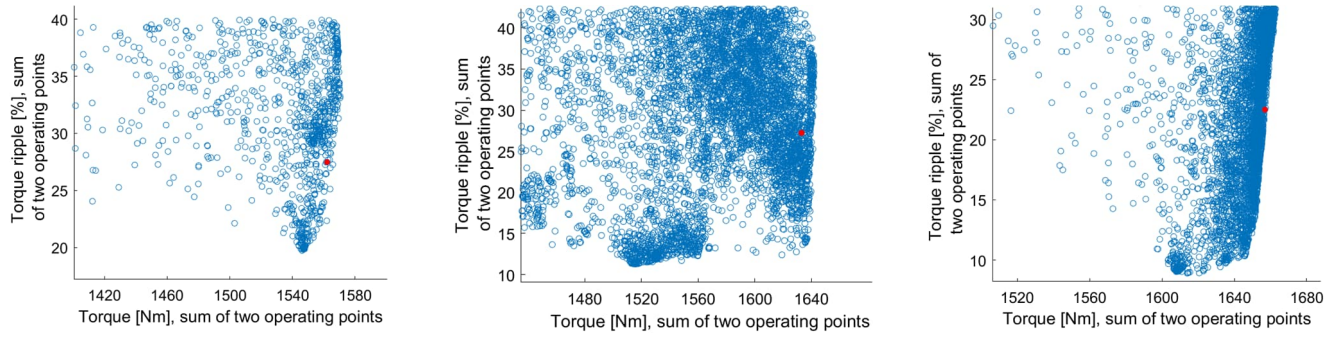
Variables	Definition
x1	Position of FG from center
x2	Thickness of FG
x3	Thickness of FG arm
x4	Width of FG body
x5	FG angle from q axis
v1	FG tooth tip arc
v2	FG tooth tip width

All results are sum of two operating points and the effective rotor mass refers to the part of rotor that is not directly supported by the lamination assembled on the shaft. Results show that the rotor with 4 FGs can create higher torque than the other two rotors but at the expense of higher effective rotor mass. Rotor with 2 FGs creates a lower torque (roughly 9% lower) but with significantly lower effective rotor mass (roughly 16% or 1 kg less effective rotor mass than with 4 FGs). Hence, the rotor with 2 FGs would, after mechanical studies, need thinner ribs in comparison with the rotor with 4 FGs where thicker ribs reduce the torque and power factor. Consequently, in the case of low effective rotor mass, the thinner ribs can effectively compensate for the otherwise reduced torque

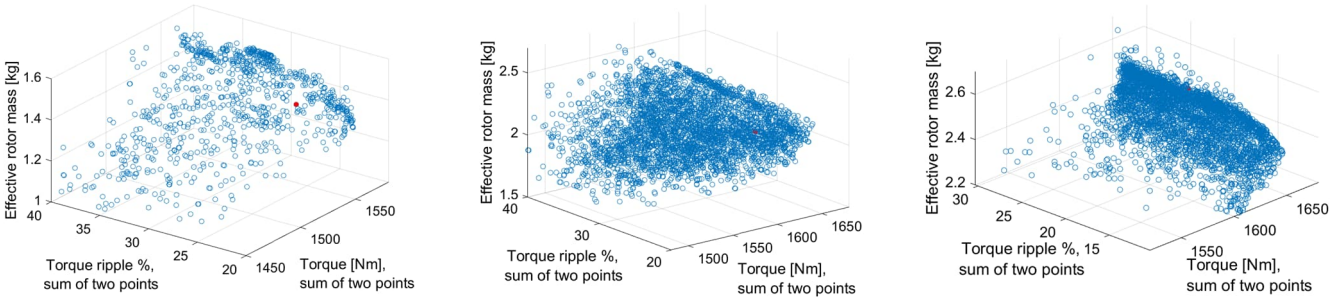
Furthermore, 3 FGs can create high torque but still need higher effective rotor mass (roughly 50% or 0.6 kg higher than 2 FGs). Consequently, tradeoffs between average torque and effective rotor mass are considerable and make significant difference in performance. Investigation of these tradeoffs and revealing the most optimized rotor geometry is the main goal in this study. In Fig. 3, three rotors are chosen (red dots) as optimal designs based on maximum torque with roughly 25% torque ripple at two operating points while having minimum rotor effective mass.



(a) Torque versus effective rotor mass (sum of two operating points) and selected designs (red dots)

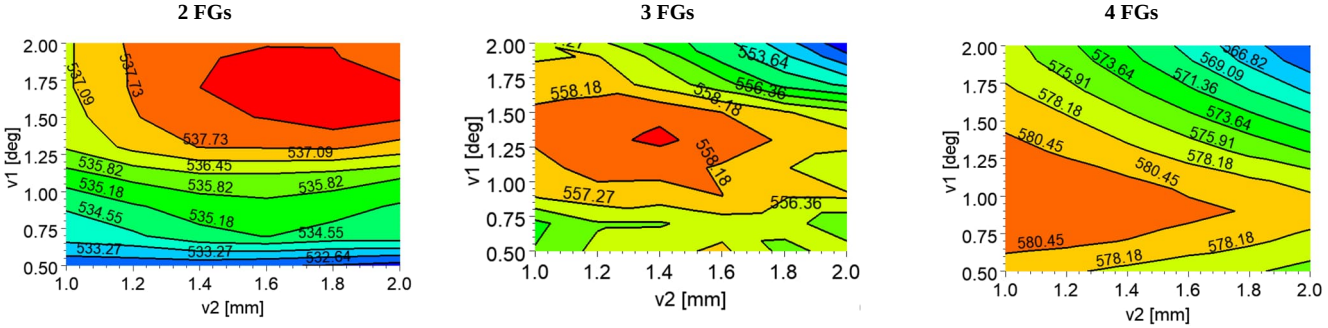


(b) Torque versus torque ripple (sum of two operating points) and selected designs (red dots)

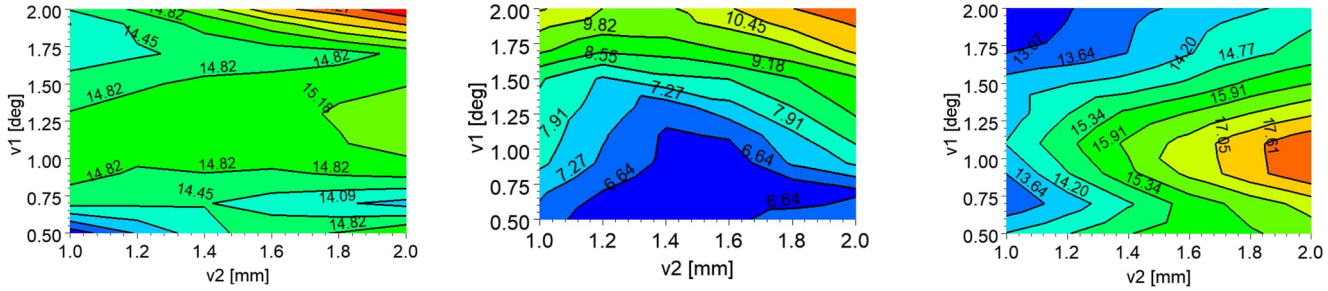


(c) 3D map of torque versus torque ripple vs effective rotor mass (sum of two operating points) and selected designs (red dots)

Fig. 3, Optimization results of three investigated rotors and the selected designs in case of ideal rotor (w/o ribs and w/o backplates).



(a) Average torque (Nm) vs. FG tooth tip dimensions (v1 and v2) for three rotors

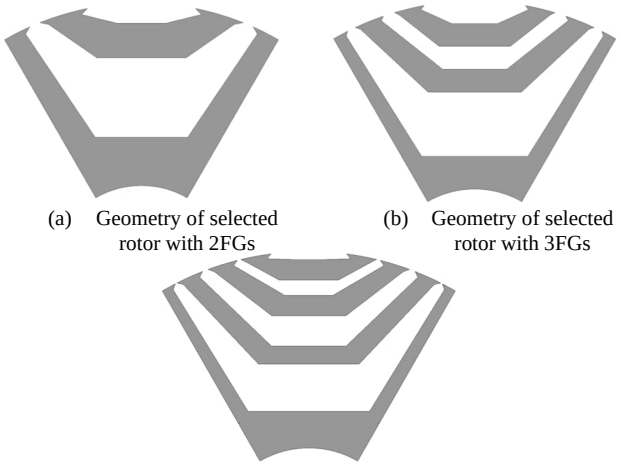


(b) Torque ripple (%) vs. FG tooth tip dimensions (v1 and v2) for three rotors

Fig. 4, Effect of FG tooth tips on average torque and torque ripple at 450 A current with 60 degrees current angle (rated current)

The rotor with 4 FGs is shown to have high torque ripple at the highest torques. This is the reason for the fact that the selected point is not on the edge of torque versus effective rotor mass. Selected designs have effective rotor mass of 1.45 kg, 2.02 kg, 2.43 kg for 2 FG, 3 FG, and 4 FG structures respectively.

After selecting the optimized design, parametric analysis is performed to determine the dimensions of FG tooth tips. It was presented in a previous publication that FG tooth tips can make significant difference in torque and torque ripple [23]. The effect of FG tooth tips on torque and torque ripple are presented for all three rotors in Fig. 4 (geometry of FG tooth tips are shown in Fig. 2). It should be noted that the dimensions of FG tooth tips can also be added to the initial optimization. However, there are high number of variables for optimization in rotors and therefore, for simplicity and also presenting the effect of FG tooth tips, this step is done separately.



(a) Geometry of selected rotor with 2FGs (b) Geometry of selected rotor with 3FGs

(c) Geometry of selected rotor with 4FGs

Fig. 5, Selected rotors (2 FGs, 3 FGs and 4 FGs) from optimization results after adjusting the FG tooth-tip dimensions

Fig. 4 shows a contour map of average torque and torque ripple versus FG tooth tip arc (v1) and FG tooth tip width (v2) as illustrated in Fig. 2 at current 450 Arms and current vector angle 60 degrees. Results in Fig. 4 demonstrate the significance of adjusting FG tooth tips to achieve lower torque ripple without compromising average torque.

According to results in Fig. 4, the best FG tooth-tip dimensions are:

- ✓ v1 = 1.6 deg, v2 = 1.8 mm for 2 FGs rotor
- ✓ v1 = 0.8 deg, v2 = 1.3 mm for 3 FGs rotor
- ✓ v1 = 1 deg, v2 = 1.1 mm for 4 FGs rotor

Fig. 5 shows the final rotors with adjusted FG tooth tips. Naturally, the tooth tips can be adjusted only in the rotors with backplates. The tangential ribs on the traditional transversally laminated rotors mostly mitigate the effect of this analysis.

TABLE III
Performance comparison of the selected designs in ideal rotor case (without ribs or backplates) with 2D FEA

	Performance at 3000 r/min	2 FGs	3 FGs	4 FGs
	Effective rotor mass (kg)	1.34	2.02	2.42
450 Arms current, 60-degree, current vector angle	Torque (Nm) and Torque ripple (%)	538.8 Nm, 14.9	556.2 Nm, 6.8 %	582.2 Nm, 14.6 %
	Output power	169.3 kW	174.7 kW	182.9 kW
	Power factor	0.62	0.65	0.65
	Copper loss (100° C), Core loss	3.63 kW, 3.66 kW	3.63 kW, 3.12 kW	3.63 kW, 3.14 kW
	L_d, L_q	881.3, 189.8 mH	895.6, 182.1 mH	937.1, 189.4 mH
	Saliency	4.64	4.91	4.94
	Efficiency (%) *	95.4 %	95.8 %	96 %
900 Arms current, 60-degree, current vector angle	Torque (Nm) and Torque ripple (%)	1025 Nm, 15.2 %	1048.1 Nm, 7.7 %	1071.8 Nm, 10.4 %
	Output power	322.1 kW	329.2 kW	336.7 kW
	Power factor	0.49	0.51	0.51
	Copper loss, (100° C), Core loss	14.54 kW, 5.1 kW	14.54 kW, 4.38 kW	14.54 kW, 4.42 kW
	L_d, L_q	492.2, 163.2 mH	494.6, 158.5 mH	526.8, 162.8 mH
	Saliency	3.01	3.12	3.23
	Efficiency (%) *	93.8 %	94.1 %	94.2 %

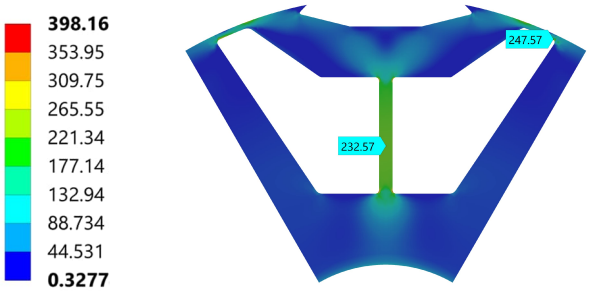
(* 0.5 % additional loss is assumed)

Table III shows the details of performance for the selected three rotors (Fig. 5) at 3000 r/min and two operating points. The results show that the rotor with 2 FGs has significantly lower effective rotor mass. However, a bit lower torque, power, power factor, and higher core loss are seen. Since the existence of ribs will significantly affect the performance of the machines, the next section provides mechanical stress analysis to calculate the thickness of ribs for three rotors to reach maximum speed of 12,000 r/min.

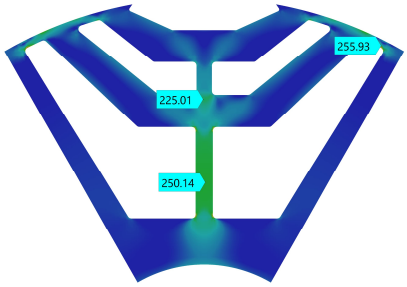
IV. MECHANICAL DESIGN OF THE CONVENTIONAL RIBS TO REACH MAXIMUM SPEED

In this section the thickness of radial and tangential ribs are calculated based on mechanical stress analysis to reach the maximum speed of 12,000 r/min. The rotor diameter is 207.8 mm. With this rotor diameter and a rotational speed of 12,000 r/min, the peripheral speed is approximately 130. 6 m/s. This high value highlights the significant challenge of maintaining the integrity of the rotor assembly against centrifugal forces. The thicknesses of radial and tangential ribs are calculated in several iterations so that the equivalent stress is roughly 250 MPa.

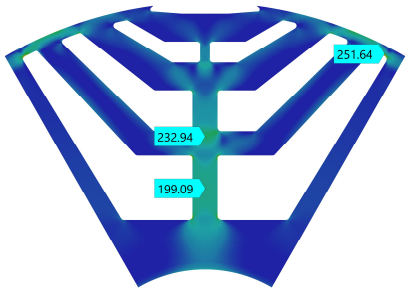
The calculated widths of the radial and tangential ribs are presented in Table IV based on stress analysis shown in Fig. 6. The radial ribs are numbered sequentially from 1 to 3, beginning with the innermost rib. Fig. 6 illustrates the stress levels for three rotors, each of which exhibits stress levels below 250 MPa.



(a) Rotor stress with 2 FGs with radial and tangential ribs



(b) Rotor stress with 3 FGs with radial and tangential ribs [MPa]



(c) Rotor stress with 4 FGs with radial and tangential ribs [MPa]

Fig. 6, Stress analysis of the three rotors at 12,000 r/min, in MPa

TABLE IV, Width of radial and tangential ribs calculated from stress analysis

Rotor type	2 FGs	3 FGs	4 FGs
Width of rib1 [mm]	3.5	4.5	6.5
Width of rib2 [mm]	--	4	6.5
Width of rib3 [mm]	--	--	2.5
Width of tangential ribs [mm]	1.5	2	2.5

The yield strength of the structural steel S355 which is used in rotor is 355 MPa. It should be mentioned that temperature dependency of S355 yield strength is studied and measured in [24]. Measurements in [24] reveal that the yield strength of S355 drops after 400°C. Hence, the stress analysis results presented in Fig. 6 remain valid because the temperature in the stator cannot rise beyond 150°C due to winding insulation temperature tolerances. Consequently, the rotor temperature will be far less than 400°C. The effect of these ribs on the performance of the machines is presented and compared in section V along with the effect of using backplates using 3D FEA.

V. PERFORMANCE COMPARISON OF ROTORS WITH CONVENTIONAL LAMINATION AND BACKPLATES WITH IDEAL CASE

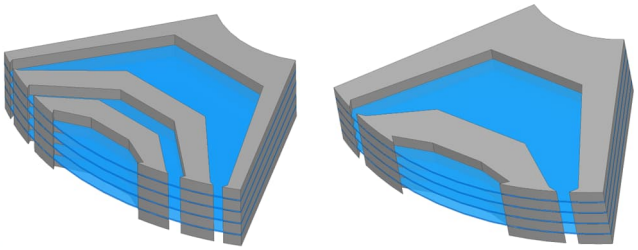
To investigate the effect of using backplates in the rotor, 3D FEA simulation is implemented, and results are calculated for three rotors with 2, 3 and 4 FGs. There are in total 47 modules of magnetic steel with thickness of 4mm each in axial length of the motor and the thickness of nonmagnetic backplates are 0.5 mm. Consequently, in this configuration the rotor steel space factor is 88.9 %. Fig. 7 shows the structure of the rotor with backplates for the first five modules and four backplates for rotors with 2 FGs and 3 FGs respectively.

The effect of using backplates on torque and power factor are summarized in Fig. 8. Ideal case refers to a rotor without ribs or backplate. The results show that using backplates reduces the torque by roughly 6 % at 450 Arms current and by 11 % at 900 Arms in comparison with the ideal case. Furthermore, PF reduces roughly by 12 % to 15 %.

It should be noted that the mechanical stress analysis of the rotor with 3 FGs using backplates (Fig. 7a) is provided in previous publication [23].

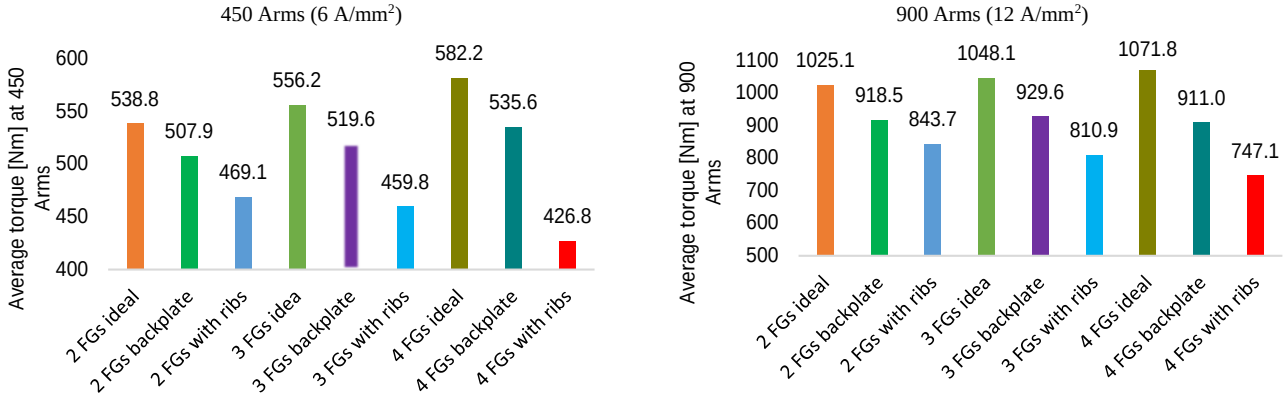
The performance indices such as torque, power factor and efficiency are presented in Fig. 8 for nine different rotors at currents of 450 Arms and 900 Arms with the same color bar for every rotor. Since stator dimensions and windings are identical in all designs, the amount of copper loss is assumed to be identical in all designs, 3.63 kW and 14.54 kW for current 450 Arms and 900 Arms respectively.

It should be noted that the calculated values in Fig. 8 are results of 3D FEA analysis because the effect of using backplates cannot be seen in 2D FEA in detail.

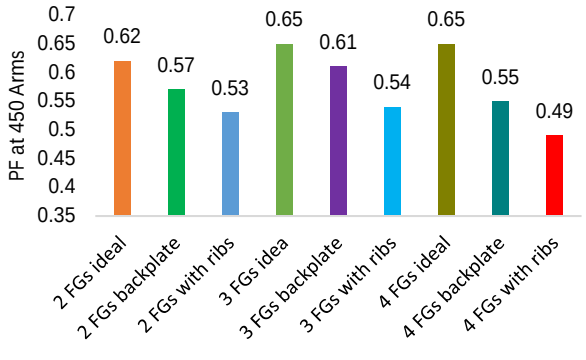


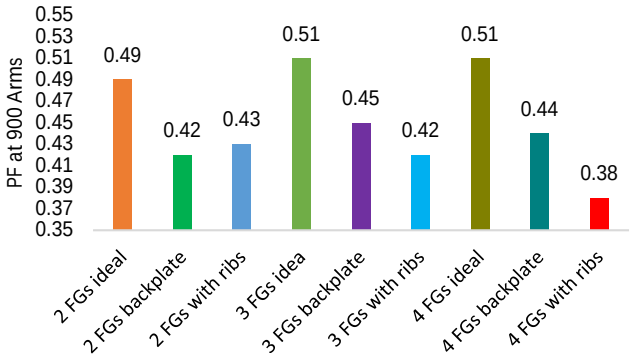
(a) 3 FGs with backplates (b) 2 FGs with backplates

Fig. 7, Rotors with backplates

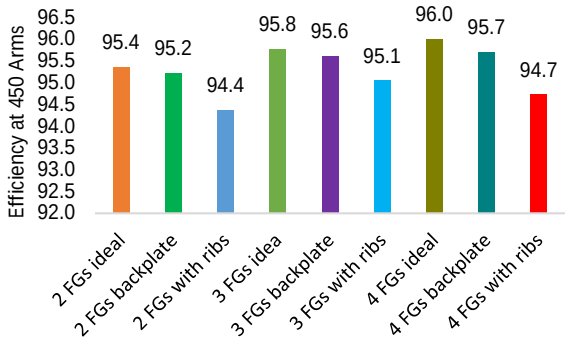


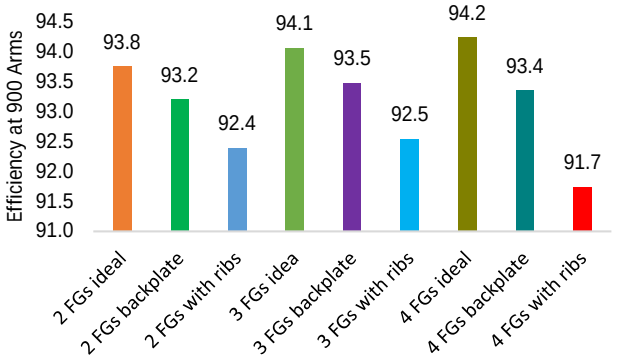
(a) Average torque at 450 Arms and 900 Arms and current vector angle 60° using 3D FEA at 3000 r/min





(b) Power factor at 450 Arms and 900 Arms and current vector angle 60º using 3D FEA at 3000 r/min





(c) Efficiency at 450 Arms and 900 Arms and current vector angle 60º using 3D FEA at 3000 r/min

Fig. 8, Comparison of Performance for three rotors, ideal case rotor (w/o ribs and no backplates) vs. rotor with ribs vs. rotor with backplate

TABLE V, Comparison of nine rotors based on specific power density (kW/kg) at 4500 rpm 900 Arms

Rotor type	Rotor total mass[1] [kg]	Total motor mass[2] [kg]	kW/kg at 900 Arms[3]
2 FGs ideal	4.34	152.1	3.175
2 FG backplate	7.56	155.4	2.786
2 FG with ribs	4.54	152.3	2.610
3 FGs ideal	3.93	151.7	3.255
3 FG backplate	8.17	155.9	2.809
3 FG with ribs	4.23	152.1	2.513
4 FGs ideal	4.21	152.1	3.322
4 FG backplate	8.52	156.3	2.746
4 FG with ribs	4.61	152.4	2.310

[1] Mass of backplates included in case of using backplates

[2] Total mass = stator housing + stator lamination + windings + rotor

[3] Torque values to calculate power are taken from Fig. 8a.

The results in Fig. 8 show that adding radial and tangential ribs significantly reduces the saliency of the rotor, torque, power factor and the efficiency in all cases, while rotors with 2 FGs and 3 FGs perform much better than the rotor with 4 FGs because the thickness of the ribs increases with the effective mass of the rotor.

Moreover, results in Fig. 8a demonstrate that, in the case of torque, rotors using backplates perform better than rotors using ribs. Fig. 8b shows that using backplates has a smaller adverse effect on the power factor than using ribs to support the flux guides.

Fig.8c shows that rotors with backplates can reach an efficiency value close to ideal rotors. For EV applications, reaching high efficiencies is the most important part since it directly affects the vehicle range.

Moreover, specific power densities of nine rotors are summarized in Table V at the speed of 4500 rpm and current 900 Arms. The masses of stator housing, stator lamination, windings are 34 kg, 81.5 kg and 32.3 kg respectively (total 147.8 kg). It should be mentioned that in the case of rotors with backplates, the mass of all backplates is calculated to be 6.37 kg and added to the rotor total mass shown in Table V. It is obvious that ideal rotors would have the highest values of specific power. The rotor with 3 FGs and backplates has the closest value to the ideal case while the rotor with 2 FGs with backplates provides only 0.82 % lower specific power than the rotor with 3 FGs and backplates.

Consequently, the results in Fig. 8 and Table V demonstrate the superiority of using the backplate technology over conventional ribs.

3D FEA results show that the rotors with backplates using 2 FGs or 3 FGs can create torques closer to the ideal case where no ribs are used, as low as possible reductions in torque and power factor is seen in comparison with ideal case.

Among rotors with backplates, the rotor with 3 FGs is selected as the optimized design due to several reasons:

- ✓ Higher torque than with 2 FGs
- ✓ Higher efficiency than with 2 FGs
- ✓ Higher PF than with 2 FGs or 4 FGs
- ✓ Torque and efficiency with 3 FGs is as high as with 4 FGs but with simpler rotor
- ✓ Closest specific power density to ideal case

In summary, according to Fig. 8, the proposed design, which

has 3 FGs with backplates delivers 13% and 15% higher torque, 13% and 7% higher power factor, 0.5% and 1% higher efficiency than 3 FG with ribs for the same currents of 450 Arms and 900 Arms respectively at the speed of 3000 r/min. Copper loss is identical in all machines since stator windings are identical.

Consequently, for a traction application, a rotor with backplate technology would require less current to deliver the same torque which would mean higher efficiencies can be achieved. Higher efficiency would reduce the energy consumption of a vehicle to travel the same distance, reducing the battery capacity needed.

It should be noted that one of the challenges in developing backplate technology is the rotor manufacturing and prototyping. Hence, lower number of FGs is beneficial since it makes rotor construction cheaper, easier and faster. Since the rotor with 3 FGs delivers higher torque and efficiency and considerably higher power factor than rotor with 2 FGs and 4 FGs, consequently, rotor with 3 FGs using backplates is selected as the optimized design among nine investigated designs in this study. However, it also can be claimed that the rotor with 2 FGs can deliver an acceptable performance and can be considered when construction needs to be simpler and faster.

The efficiency map (Fig. 9) of rotor with 3 FGs is created considering 800 Arms current (10.74 A/mm^2) limit and voltage limit of 450 Vrms line to line (10% voltage reserve assumed) while considering the maximum torque per current (MTPA) control method. Since 3D FEA requires a significant amount of time, it is practically impossible to simulate all points needed to create an efficiency map with 3D FEA. Hence, the efficiency map of the selected rotor with 3 FGs and backplates is generated using 2D FEA while reducing the active length of motor from

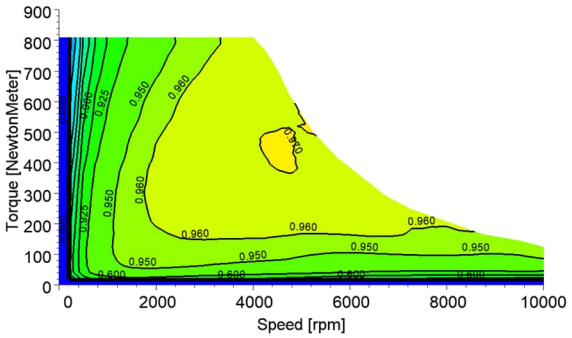


Fig. 9, Efficiency map of the SynRM with the rotor with 3 FGs and backplates

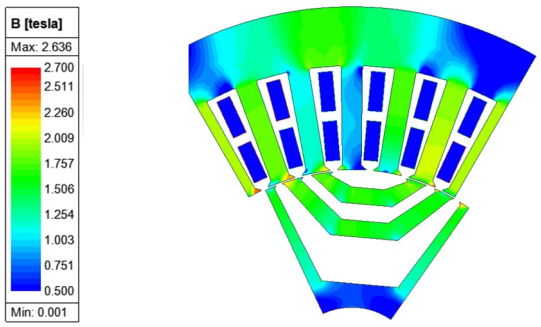


Fig. 10, Flux density distribution at 800 Arms

210 mm to 178.5 mm (15 % from Fig. 8, 10% reduction due to backplate and 5 % reduction due to implemented skew) to compensate for the effect of backplates while considering the worst-case scenarios.

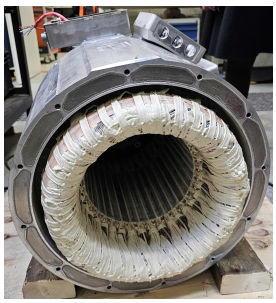
Phase resistance is calculated to be 4.85 mΩ at 100º C. The mechanical and AC winding losses are ignored in the efficiency map shown in Fig. 9. At peak current of 800 Arms this motor can reach 4000 r/min and produce 800 Nm torque, i.e. 340 kW power, and at continuous current of 450 Arms, this motor can produce roughly 260 kW at 5000 r/min.

The flux density of the rotor with 3 FGs at 800 Arms is demonstrated in Fig. 10 to complete the study for the rotor with 3 FGs.

VI. PROTOTYPING AND MEASUREMENTS

The 3 FG rotor, optimized and compared in section III, has a simpler structure for backplate manufacturing and achieves performance and efficiency close to an ideal rotor without ribs. Hence, it was decided to manufacture a SynR motor with the rotor having 3 FGs. The manufactured stator is shown on Fig. 11a. The stator dimensions are described in Table I. Stator has round wire short pitch (5/6) windings with 20 turns/phase.

Rotor manufacture has been done in several steps. First, rotor laminations are cut according to 3 FG design as shown in Fig. 5 and Fig. 11b. Then four laminations are spot-welded together to create the 4 mm module, Fig. 11c. Then the backplates are attached to the rotor modules to complete it as shown in Fig. 11d. The final steps are to attach the rotor modules with backplates together on the shaft and also perform 10-degree mechanical continuous skew to complete the rotor manufacture as shown in Fig 11e. Finally, the tangential ribs seen in Fig. 11b are milled away. The final rotor is shown on Fig. 11f. As seen in Fig.11f deformations are seen after milling. Hence, a better manufacturing method is under investigation to simplify and accelerate the rotor manufacture.



(a) Manufactured stator



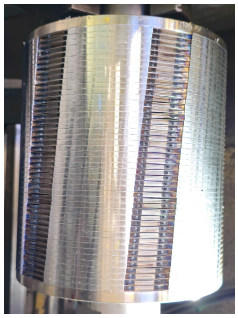
(b) One rotor lamination



(c) One rotor module without backplate



(d) Rotor module after attaching backplate

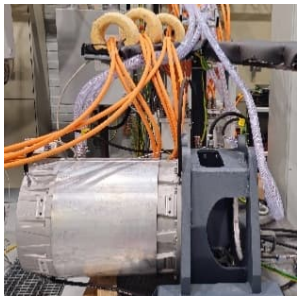


(e) Rotor after attaching all modules together

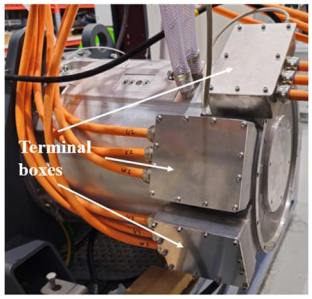


(f) Final rotor after milling away the outer ribs

Fig. 11, Manufacture of the proposed SynR motor



(a) Back-to-back test bench



(b) Installed motor



(c) ABB converter with peak current up to 900 A

Fig. 12, Installed equipment for the measurement in the laboratory

TABLE VI, Mechanical properties of lamination and backplate materials

Material	316L	S355
Elastic modulus [GPa]	193	200
Yield strength [MPa]	290	355
Density [kg/m3]	8000	7850
Poisson's ratio	0.3	0.3
Thermal Expansion Coefficient	16.0×10^{-6}	12.0×10^{-6}

The rotor backplate material is austenitic stainless steel 316L with thickness of 0.5mm which has a yield strength of 290 MPa. In case of higher stress stronger material can be selected. In this case, thickness of backplate is investigated, and minimum thickness (0.5mm) is selected to ensure both mechanically strong structure while achieving highest possible rotor space factor.

Moreover, the yield strength of stainless steel 316L base material reduces to roughly 220 MPa at 100°C from 290 MPa at 25°C [24]. In this study the rotor operating temperature is assumed to be 100°C.

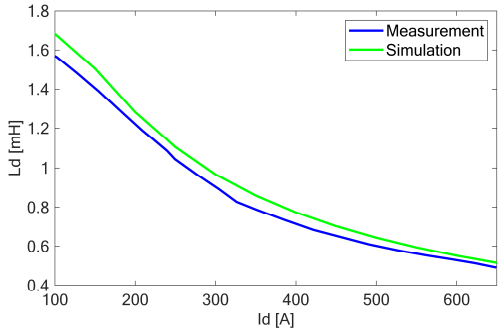
The existence of two different materials in the rotor might create mechanical issues due to their different properties such as thermal expansion, yield strength, elasticity modulus etc. The mechanical properties of S355 and 316L are summarized in Table VI. At high temperatures the rotor will elongate in the axial direction and its diameter will increase both with temperature and rotational speed. Hence, certain measures are considered to have sufficient clearance between rotor and other parts. The higher thermal expansion of 316L might affect rotor stiffness and rotor dynamics. Rotor dynamics and mechanical analysis results are recently published in IEMDC 2025 [25]. It should be noted that the rotor has been rotated up to 4000 r/min.

A back-to-back test is planned to measure the performance of the machine by applying ABB's HES 880-902 inverters to control the motors as soon as the test bench and all equipment are available as shown in Fig. 12. Fig. 12b shows three terminal boxes of the motor, each terminal box provides currents to two parallel paths that make a total of six parallel paths for the motor.

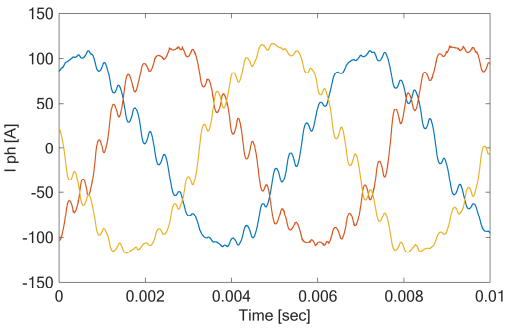
Currently, initial no load runs have been completed and phase resistance measured to be 3.75 mΩ at room temperature.

Fig. 13a demonstrates the d-axis inductance from measurements and FEA 2D simulations. Measurement and simulation results are close. Flux leakages in actual motor causes the L_d value from measurements to drop slightly in comparison with the simulation results. Fig.13b shows the phase current at no load at 3000 r/min.

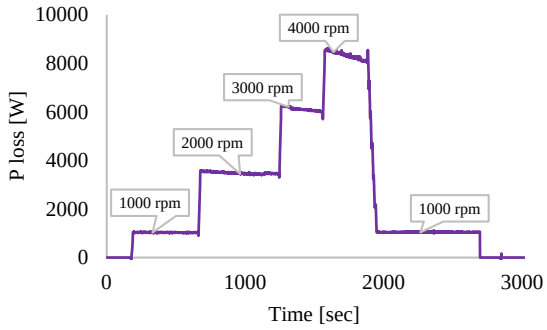
Fig. 13c shows the no load power loss to be 1000 W, 3500 W, 6000 W and 8000 W at 1000 r/min, 2000 r/min, 3000 r/min and 4000 r/min respectively. The ABB converter tries to adjust the current vector amplitude and angle to increase the efficiency of the motor. This is the reason the current amplitude reduces slightly through time in Fig.13c, mostly seen at 3000 r/min and 4000 r/min. Fig.13d shows the temperature level for different parts of motor with the same speed envelope seen in Fig.13c.



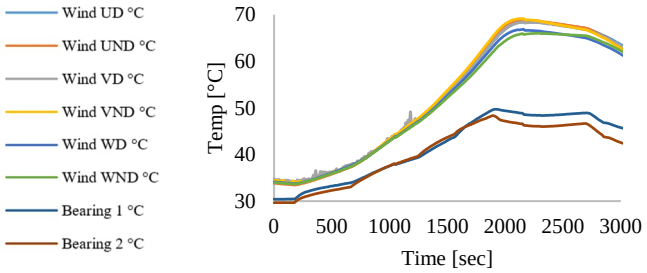
(a) L_d, d-axis inductance measurement and simulation vs. Id



(b) No-load phase current at 3000 rpm with DTC by ABB HES



(c) No load power losses up to 4000 rpm speed with converter control (3 kHz switching frequency)



(d) Temperature levels in noload operation in same speed envelope at (c)

Fig. 13, Measurement results

The no-load loss is higher than calculated. This is a result of difficulties in the rotor manufacturing. The original idea was to insulate the rotor modules from each other but that was not successful in the first prototype. In addition, the rotor was manufactured using constructional 1 mm thick S255 laminations and the airgap harmonics create significant losses in this material. In future work, low-loss electrical steel laminations will be used also in the rotor construction.

VII. Conclusion

A comprehensive optimization methodology for design of high-speed synchronous reluctance motors presented. The proposed design methodology comprised of several steps. Firstly, electromagnetic optimization is carried out based on the introduced geometry variables for rotor with two, three and four FGs. Secondly, flux-guide tooth tips are adjusted with parametric analysis. Lastly, mechanical analysis is utilized to calculate the thickness of ribs in case of conventional rotor. Effective rotor mass was introduced and utilized as an objective to minimize the mass of rotor pole. This objective leads to lower centrifugal forces on the rotor support arrangements that can be either radial and tangential ribs or backplates. Therefore, in identical condition a rotor with lighter pole mass can achieve higher speed or from other point of view, rotor can have thinner ribs or backplates that can enhance performance such as delivered torque, power factor and efficiency with the same input current. Furthermore, two operating points were utilized in the optimization process to consider both continuous and peak performance at the design level.

Optimization results were investigated, and three rotors were selected as optimized rotors based on torque, torque ripple and effective rotor mass. The effect of the proposed backplate technology was thoroughly discussed and compared with conventional ribs in case of optimized rotors. 3D FEA results demonstrated the superiority of using backplate technology for high speed SynRMs.

It should be mentioned that both optimized rotors with 2 FGs and 3 FGs have good performance and are good candidates for prototyping. The rotor with 4 FGs results in higher rotor mass which needs thicker ribs that have significant adverse effects on the performance. Moreover, manufacture of 4 FGs rotor with backplates requires thicker backplates due to higher rotor mass of 4 FG. Moreover, manufacture of 4 FGs is much more complicated and costly than with 2 FGs or 3 FGs. Consequently, the rotor with 3 FGs was chosen to be manufactured due to its higher torque, power factor and efficiency than with the rotor with 2 FGs and easier manufacturability in comparison with 4 FGs.

3D FEA results of rotor with 3 FGs and backplates show that with using backplates the performance of the motor can get closer to ideal case with no ribs and no backplates. It was calculated that the adverse effect of the backplates on the average torque is roughly 6 % at 450 Arms and 11 % at 900 Arms in comparison with the ideal case. Furthermore, PF reduces by roughly 12 % to 15 %. However, according to results shown in Fig. 8 using backplates enhances torque (13%~15%), PF (7%~13%) and efficiency (0.5%~1%) at same currents in comparison to using conventional ribs. Therefore, the proposed backplate technology can provide closer performance to ideal case (no ribs and no backplates) than rotors with conventional ribs. Moreover, results demonstrated that utilizing backplates significantly enhances the efficiency of SynRMs, which is important to make SynRMs feasible for traction applications. Higher efficiency can positively affect vehicle range and overall cost by completely removing expensive magnets from motor.

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Arash Allahyari (M’21) born in 1989 Urmia, Iran. received the M.Sc. degree in electrical power engineering from Urmia University in 2011. He was a research assistant at Middle East Technical University, Türkiye on design of traction motors from 2020 to 2022. Since 2023, he has been pursuing a D.Sc. degree in electrical engineering at Lappeenranta University of Technology (LUT) as a part of the EU Horizon 2020 Marie Skłodowska-Curie Actions (MSCA) Doctoral Network. His research interests include the optimization and design of synchronous reluctance and interior permanent magnet traction motors.



Ilya Petrov received the D.Sc. degree from the Lappeenranta University of Technology (LUT), Finland, in 2015. He is currently a Research Fellow of the Department of Electrical Engineering, LUT.



Juha J. Pyrhönen (M’06, SM’17) was born in 1957 in Kuusankoski, Finland. He received the Doctor of Science (D.Sc.) degree in electrical engineering from Lappeenranta University of Technology (LUT), Finland, in 1991. He was appointed Professor of Electrical Machines and Drives at LUT in 1997, where he led research and education in the field for nearly three decades. Throughout his career, Prof. Pyrhönen has contributed extensively to the development of high-performance electric drives and machines, with applications ranging from distributed energy generation to traction and high-speed systems. His research has placed particular emphasis on permanent magnet machines—both in terms of optimal application and alternatives to rare-earth materials. More recently, his work has explored the integration of carbon-based materials in electrical machine design. Prof. Pyrhönen retired in 2025 and continues his academic involvement as Professor Emeritus at LUT.



Juho Montonen (M’12, SM’18) was born in Joutseno, Finland, in 1988. He received M.Sc. degrees and D. Sc. (Tech.) degrees from the Lappeenranta University of Technology (LUT), Finland, in 2012 and 2018, respectively. He works as a Chief Engineer at Danfoss Editron now. His research interests include electrical machines and drives. His research focuses on permanent magnet motors, with a special reference to different kinds of traction applications.



Lassi Aarniovuori (IEEE Senior Member) received M.Sc. degree in electrical engineering and the D.Sc. degree in electric motors and drives from the Lappeenranta University of Technology (LUT), Lappeenranta, Finland, in 2005 and 2010, respectively. During 2017 – 2019 he was a Marie Curie Fellow with the School of Engineering and Applied Science, Aston University, Birmingham, U.K. He is currently an Associate Professor in Electric Transportation with the LUT School of Energy Systems. His current research interests include all aspects related to electric transportation systems, especially wide band-gap power switches, high-speed machine technology, modulation methods, simulation of electric drives, efficiency measurements, and calorimetric measurement systems.